QUALITY MANAGEMENT SYSTEM DEVELOPMENT

Steel Fabrication – Welding – Galvanizing Operations

Project Portfolio Documentation

Developed for:

PT Bagja Mulia Technologies

1. Executive Summary

This project involved the development of a structured Quality Management System (QMS) for steel fabrication workshop operations. The system was designed to ensure that all production activities are controlled through standardized procedures, inspection checkpoints, and traceability documentation.

The implemented QMS provides a complete documentation framework covering material inspection, welding process control, dimensional verification, galvanizing inspection, and final product release.

Through this system, workshop operations are supported by clearly defined procedures, qualified personnel requirements, and inspection records that ensure product quality and process traceability.

The system also introduces documentation structures commonly used in industrial quality management environments, enabling improved operational consistency and audit readiness.

2. Company Operational Scope

PT Bagja Mulia Technologies operates in the field of steel fabrication and structural welding services. The workshop performs fabrication processes that require strict quality control to ensure structural integrity and compliance with engineering specifications.

Operational activities include:

• Raw material receiving

• Incoming material inspection

• Material cutting and preparation

• Component fit-up assembly

• Structural welding operations

• Dimensional inspection

• Surface preparation

• Hot-dip galvanizing process

• Final quality inspection and product release

Each stage of the fabrication process requires quality verification to ensure that the final product meets both engineering requirements and customer specifications.

3. Project Objectives

The primary objective of this project was to develop a comprehensive Quality Management System that can support fabrication workshop operations.

The system was designed to achieve the following objectives:

• Establish structured quality documentation for fabrication processes

• Ensure full traceability of materials and welding activities

• Implement standardized inspection checkpoints throughout production

• Control welder qualification and welding accountability

• Manage calibration status of inspection equipment

• Provide documentation suitable for quality audits and client inspections

Through these objectives, the system supports improved operational discipline and quality control reliability.

4. Quality Management System Architecture

The developed Quality Management System follows a hierarchical documentation structure commonly used in industrial quality systems.

The documentation architecture consists of four levels:

Level 1

Quality Manual

This document defines the quality management framework including policies, organizational structure, and operational quality principles.

Level 2

Standard Operating Procedures (SOP)

Operational procedures controlling fabrication activities such as material inspection, welding operations, and final quality verification.

Level 3

Quality Control Inspection Forms

Inspection documents used to record verification activities throughout the production process.

Level 4

Registers and Control Logs

Traceability records and monitoring systems used to manage welder qualification and inspection equipment calibration.

This hierarchical structure ensures that operational processes are supported by structured documentation and traceability systems.

5. Organizational Structure

The Quality Management System also defines the operational responsibilities within the fabrication workshop.

The organizational structure establishes clear roles for departments involved in fabrication operations including:

• General Management

• Engineering Department

• Production Department

• Quality Control Department

• Warehouse Department

Each department has defined responsibilities in ensuring that operational activities comply with established procedures and quality standards.

This structure supports clear communication channels and accountability across the fabrication workflow.

6. Quality Manual Development

The Quality Manual serves as the primary reference document of the Quality Management System.

This document defines the quality policy, system scope, and management responsibilities

related to fabrication operations.

Key topics covered in the manual include:

• Quality policy and system objectives

• Organizational responsibilities

• Process control requirements

• Inspection management procedures

• Non-conformance control

• Corrective and preventive actions

• Internal audit principles

The Quality Manual provides the foundation upon which operational procedures and inspection documentation are built.

7. Standard Operating Procedures Development

To ensure that fabrication operations are performed consistently and according to defined quality standards, a set of Standard Operating Procedures (SOPs) was developed as part of the Quality Management System.

These procedures define operational requirements, inspection checkpoints, and documentation requirements for each stage of fabrication.

The following SOPs were developed to control the production workflow.

Incoming Material Inspection

This procedure establishes the requirements for verifying raw materials received at the workshop.

Inspection activities include:

• Verification of material against purchase order specifications

• Visual inspection of material condition

• Heat number identification

• Verification of Mill Test Certificate (MTC)

Purpose

The objective of this procedure is to ensure that only materials that meet technical specifications are accepted for fabrication processes.

Cutting and Fit-Up Control

This procedure controls the preparation of components before welding operations.

Inspection activities include:

• Verification of cutting dimensions

• Joint preparation inspection

• Root gap measurement

• Alignment verification

Purpose

Ensuring proper joint preparation is critical for maintaining welding quality and structural integrity.

Welding Process Control

Welding operations are among the most critical processes in fabrication. This procedure establishes controls to ensure that welding activities are performed according to qualified procedures.

Controls include:

• Verification of welder qualification

• Compliance with Welding Procedure Specifications (WPS)

• Monitoring welding parameters

• Post-weld visual inspection

Purpose

This procedure ensures that welding activities are performed by qualified personnel and meet the required quality standards.

Dimensional Inspection

Dimensional verification is performed after welding operations to ensure that fabricated components comply with engineering drawings.

Inspection activities include:

• Measurement of component dimensions

• Verification of alignment and tolerances

• Identification of deviations from design specifications

Purpose

This inspection ensures that fabricated structures meet dimensional requirements before further processing.

Surface Preparation

Prior to galvanizing, fabricated components must undergo surface preparation to remove contaminants that could affect coating quality.

Inspection activities include:

• Removal of welding slag and debris

• Cleaning of oil or grease contamination

• Verification of surface cleanliness

Purpose

Proper surface preparation is essential to ensure adhesion and quality of galvanizing coating.

Galvanizing Inspection

This procedure defines inspection activities after the galvanizing process.

Inspection activities include:

• Visual inspection of coating quality

• Measurement of coating thickness

• Identification of coating defects

Purpose

Ensuring that the galvanizing layer provides adequate corrosion protection.

Final Quality Control Release

Before products are delivered to clients, a final inspection is performed to verify that all fabrication processes and inspections have been completed.

Verification includes:

• Confirmation of inspection records

• Verification of dimensional compliance

• Review of quality documentation

Purpose

This stage ensures that only products meeting quality requirements are released.

8. Quality Control Inspection Documentation

To support the inspection procedures, a set of Quality Control inspection forms was developed to document verification activities.

These forms ensure that each production stage is supported by recorded inspection evidence.

The system includes the following inspection documentation:

• Material Receiving Report

• Incoming Material Inspection Checklist

• Mill Certificate Verification Report

• Material Traceability Log

• Fit-Up Inspection Report

• Welding Inspection Report

• Dimensional Inspection Report

• Surface Preparation Checklist

• Galvanizing Thickness Inspection Report

• Final Quality Release Certificate

Each form records inspection results, responsible personnel, and verification status.

This documentation system provides traceability and accountability across fabrication operations.

9. Welder Qualification Control System

Welding operations require qualified personnel to ensure structural integrity and compliance with welding standards.

A Welder Qualification Control System was developed to manage welding personnel credentials.

The system records:

• Welder identification number

• Qualified welding processes

• Welding position qualifications

• Reference welding procedure specification

• Certification validity period

Each weld performed during fabrication can be traced to the responsible welder through identification records.

This system improves accountability and supports traceability in welding operations.

10. Equipment Calibration Management

Accurate measurement instruments are essential for reliable inspection activities.

To ensure measurement accuracy, a calibration management system was developed consisting of:

• Master Equipment Register

• Calibration Log

The Master Equipment Register records inspection instruments used within the workshop.

The Calibration Log monitors calibration schedules and ensures that equipment remains within valid calibration periods.

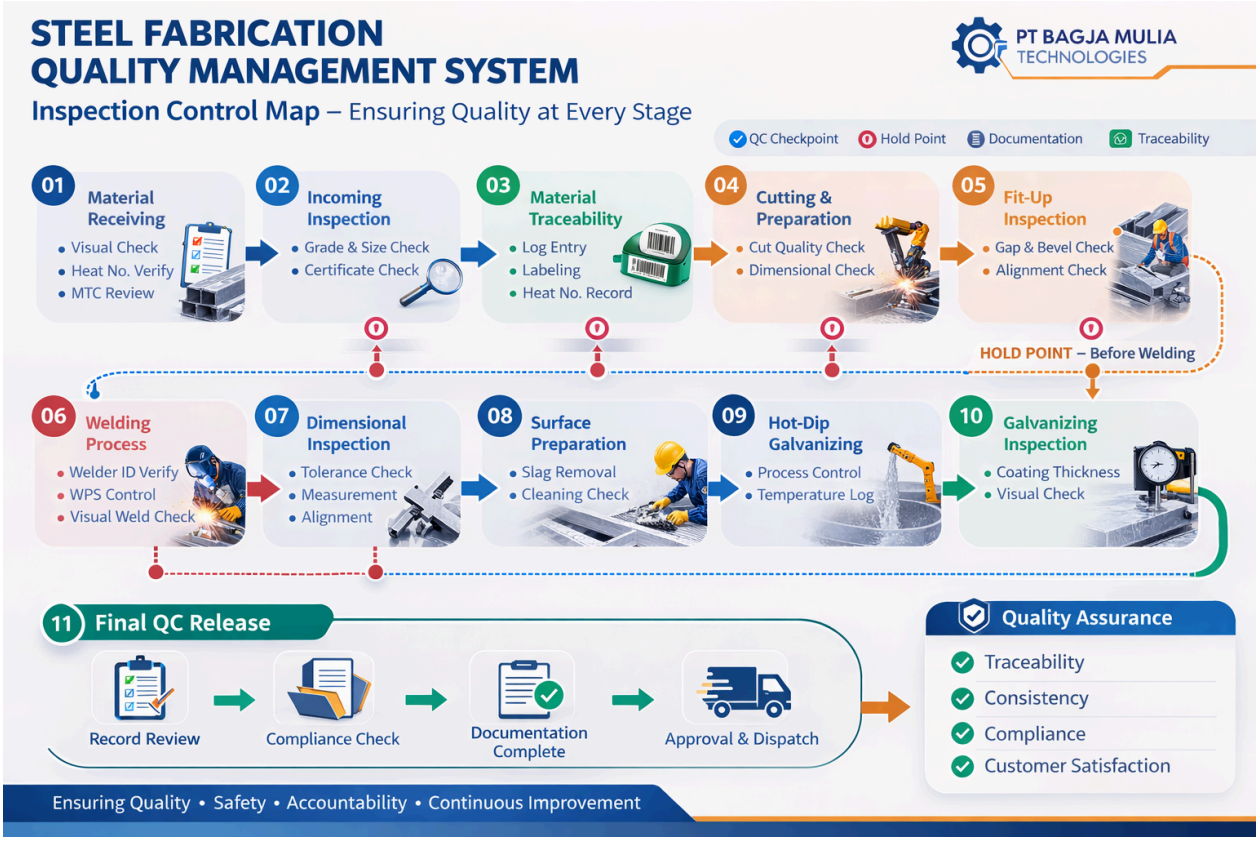
Inspection instruments with expired calibration status are restricted from use until recalibration is completed.

This system ensures the reliability of inspection measurements and supports quality assurance.

11. Fabrication Quality Control Workflow

The Quality Management System establishes inspection checkpoints across the entire fabrication workflow.

The production process follows the sequence below:



Each stage is supported by documented inspection procedures and records.

This structured workflow ensures that quality verification occurs throughout the entire production process.

12. System Benefits

The implementation of the Quality Management System provides several operational advantages.

Improved Traceability

Materials, welding operations, and inspection activities can be traced through documented records.

Standardized Inspection Process

Inspection procedures ensure consistent verification throughout fabrication operations.

Improved Quality Control

Structured inspection checkpoints reduce the risk of fabrication defects.

Audit Readiness

Documentation generated by the system supports internal audits, external audits, and client inspections.

Operational Consistency

Standardized procedures improve process discipline and operational reliability.

13. Professional Competencies Demonstrated

This project demonstrates professional capabilities in the following areas:

Quality Management System Development

Industrial Process Documentation

Quality Control Workflow Design

Welding Quality Control Systems

Material Traceability Management

Inspection Documentation Development

These competencies support the development of structured operational systems within industrial fabrication environments.